

DESIGN AND IMPLEMENTATION OF A DRIVER DROWSINESS DETECTION AND SPEED CONTROL SYSTEM FOR TRUCK DRIVERS

Presented by

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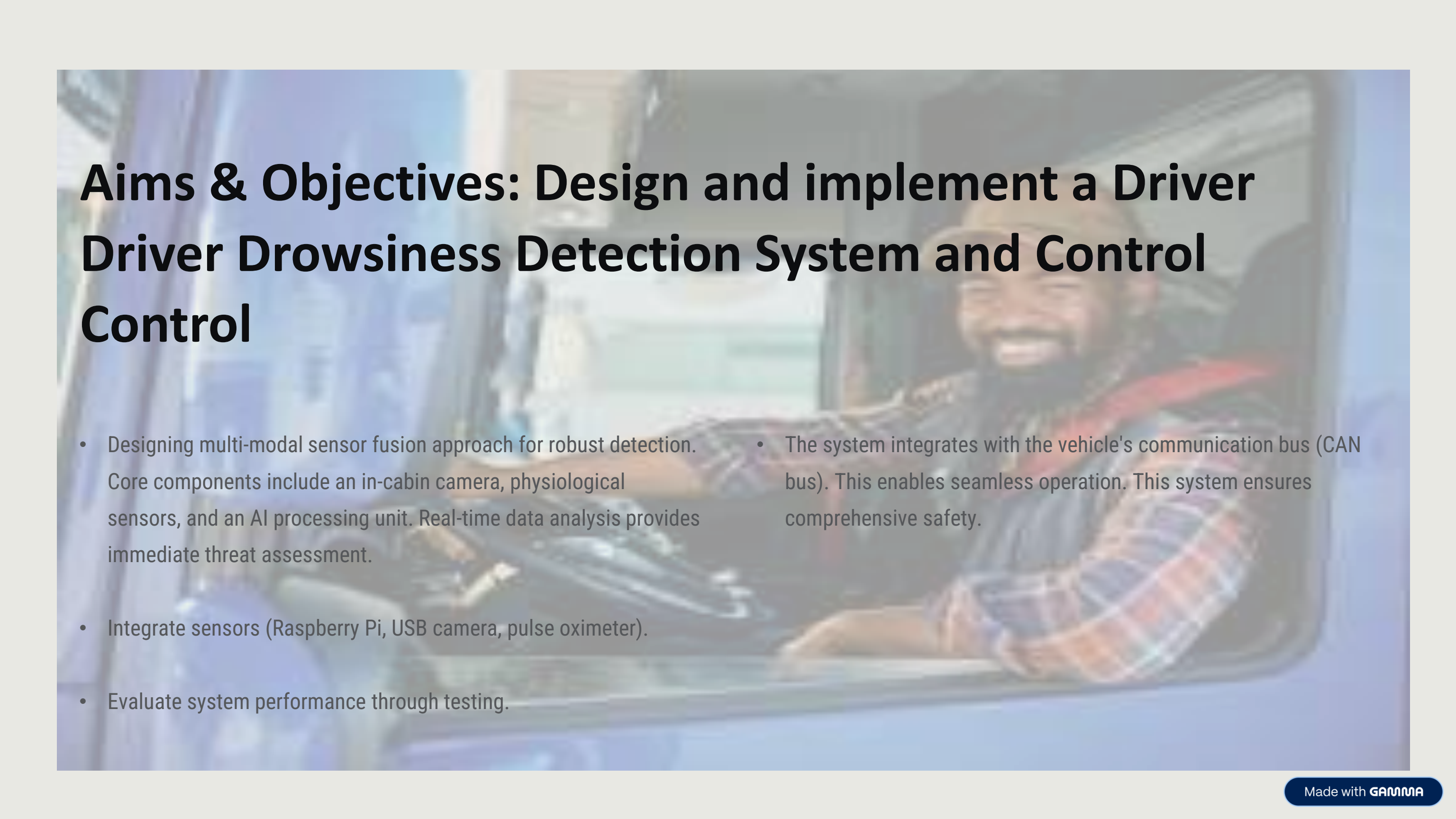
Introduction: The Silent Killer on Our Roads

- Driver fatigue is a major factor in severe global road accidents. Drowsiness impairs driving ability. It is similar with having a 0.05% blood alcohol content after 18 hours awake. Proactive systems are essential for public safety.
- Driver drowsiness is a significant contributor to road accidents globally, leading to loss of life, injuries, and economic costs



Problem Statement: The Alarming Impact of Drowsy Driving

- Current drowsiness detection methods are unreliable (subjective self-assessment).
- Existing systems often rely on single indicators (e.g., steering patterns).
- Inaccessibility of technology for average drivers.
- There is a need for a comprehensive, real-time, and cost-effective system to accurately detect drowsiness.



Aims & Objectives: Design and implement a Driver Drowsiness Detection System and Control

- Designing multi-modal sensor fusion approach for robust detection. Core components include an in-cabin camera, physiological sensors, and an AI processing unit. Real-time data analysis provides immediate threat assessment.
- Integrate sensors (Raspberry Pi, USB camera, pulse oximeter).
- Evaluate system performance through testing.
- The system integrates with the vehicle's communication bus (CAN bus). This enables seamless operation. This system ensures comprehensive safety.



Project: Advanced Detection Methods Employed



Visual Cues

Monitors eye closure (PERCLOS >80% indicates high drowsiness), blink rate, yawning, and head nodding.



Physiological Signals

Measures Heart Rate Variability (HRV), SPO Levels



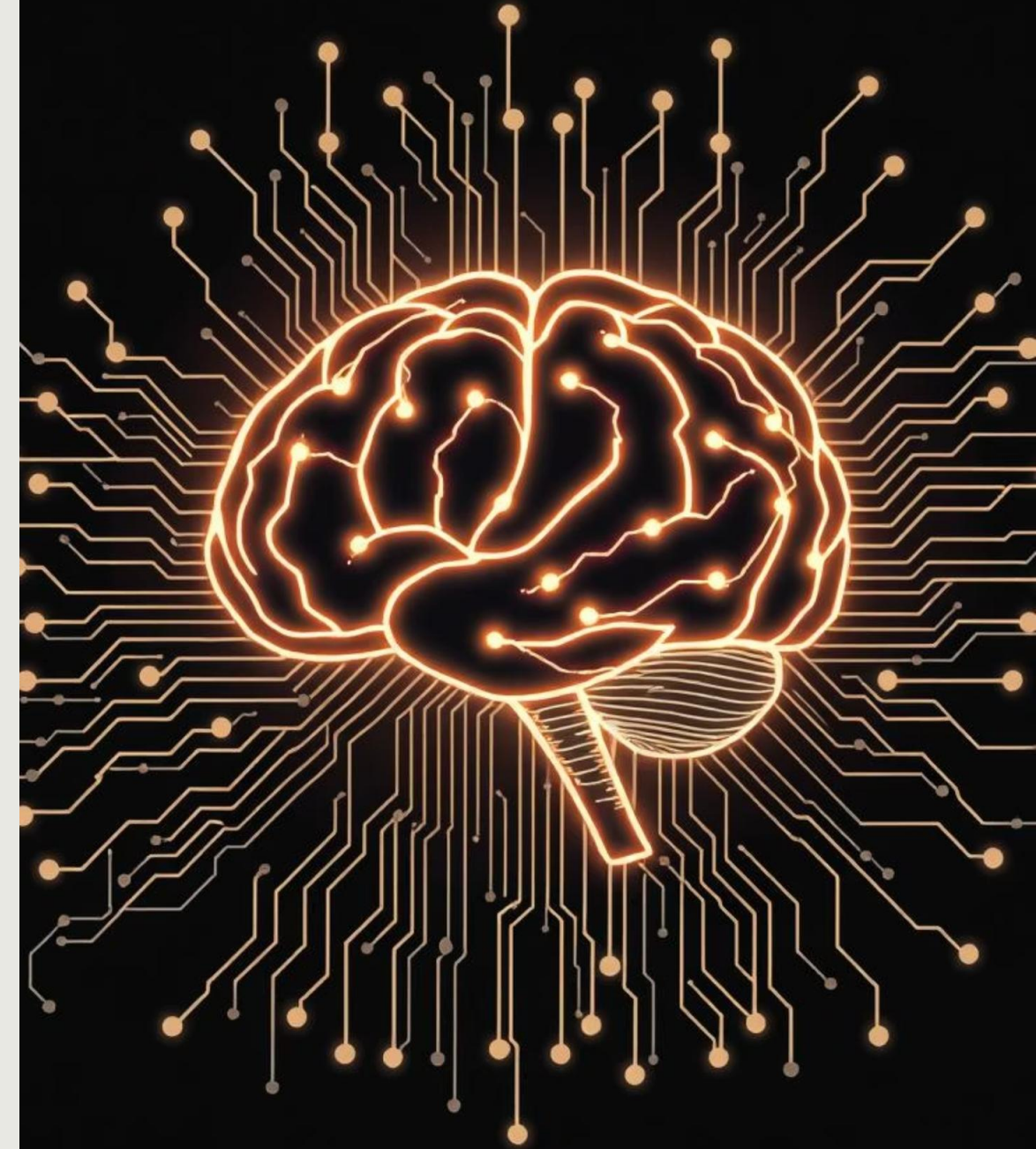
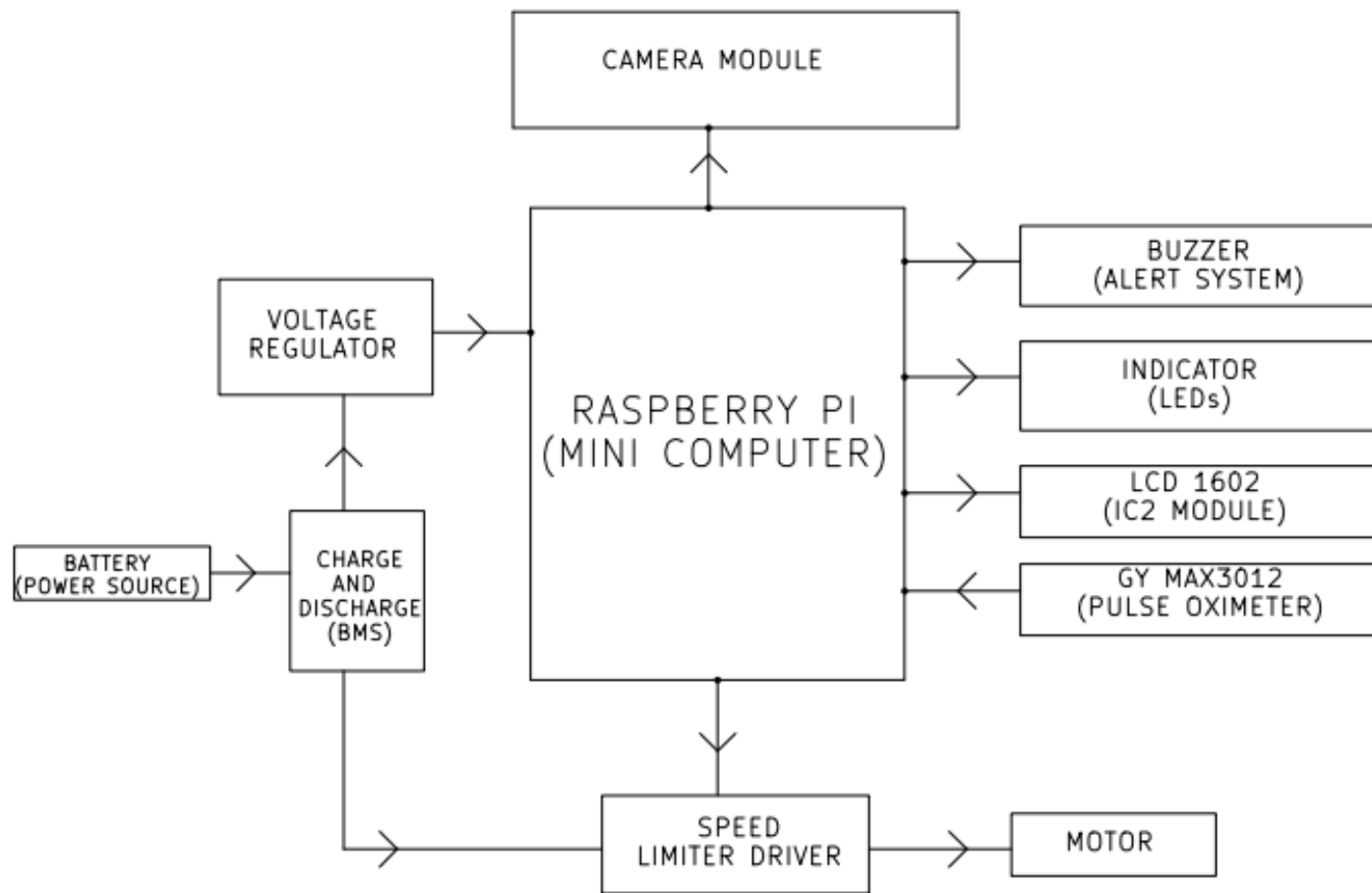
Test Algorithms

Correlates all inputs for 95%+ detection accuracy, ensuring reliability.

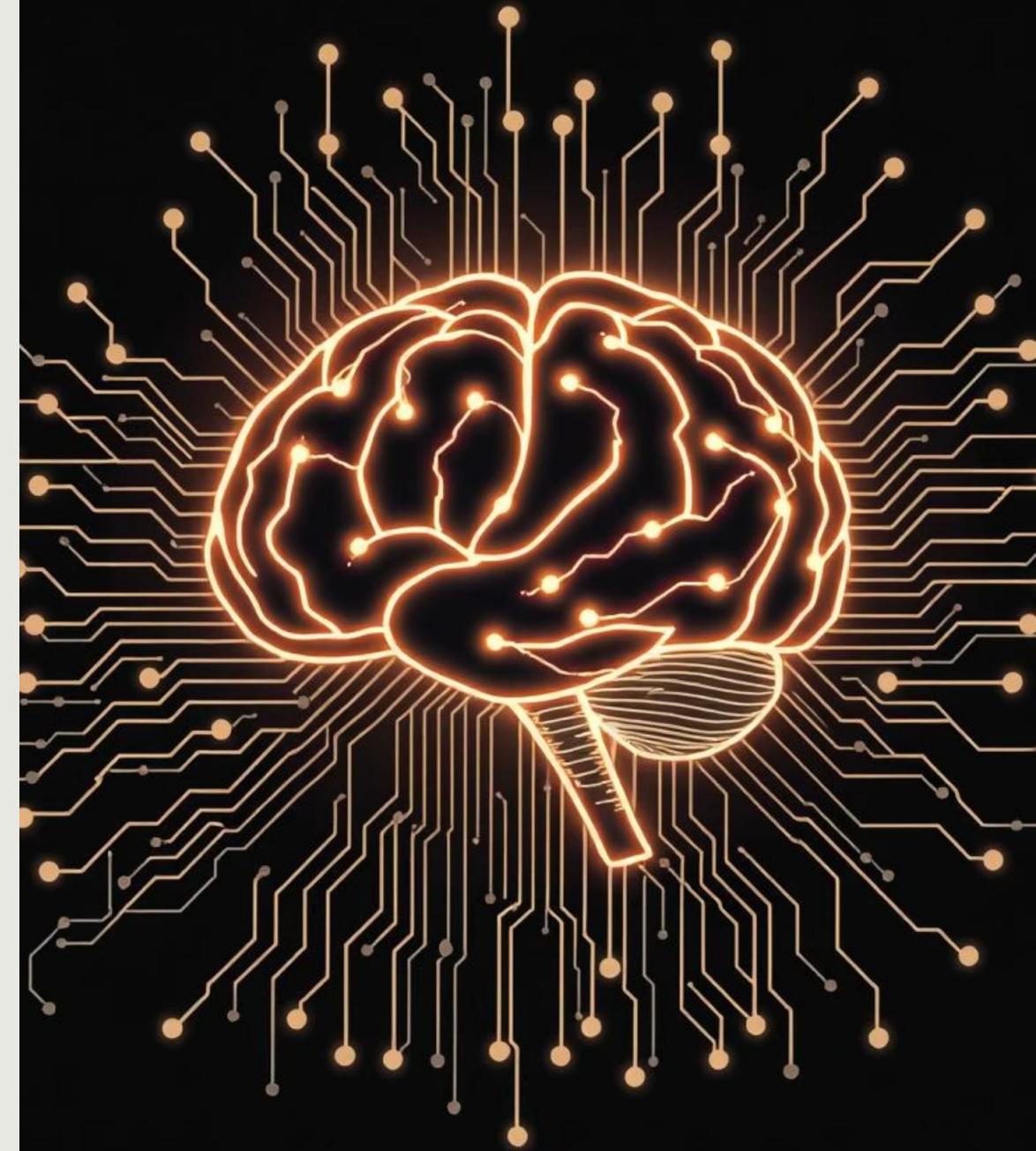
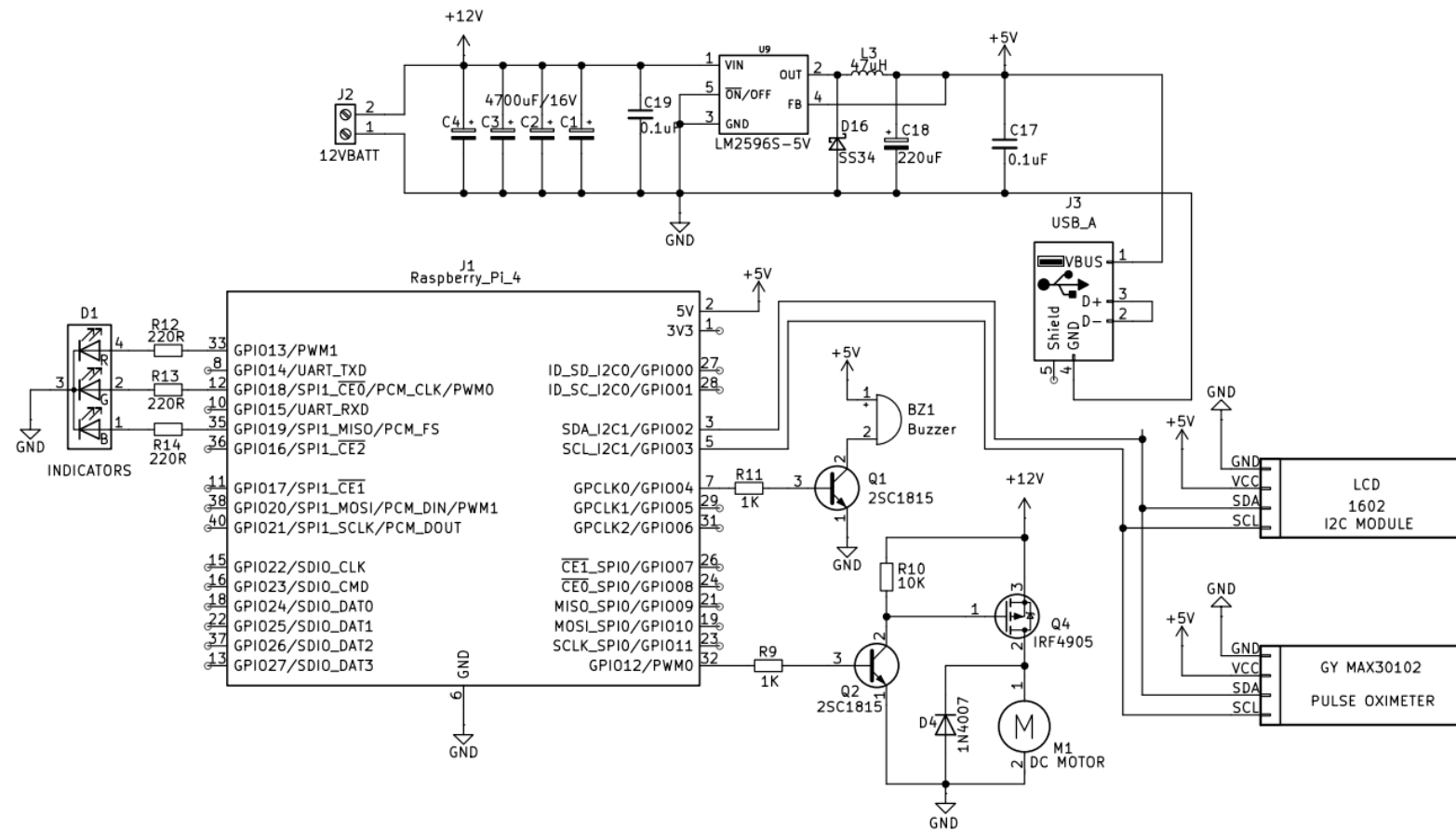
Project: Proactive Control & Mitigation Strategies

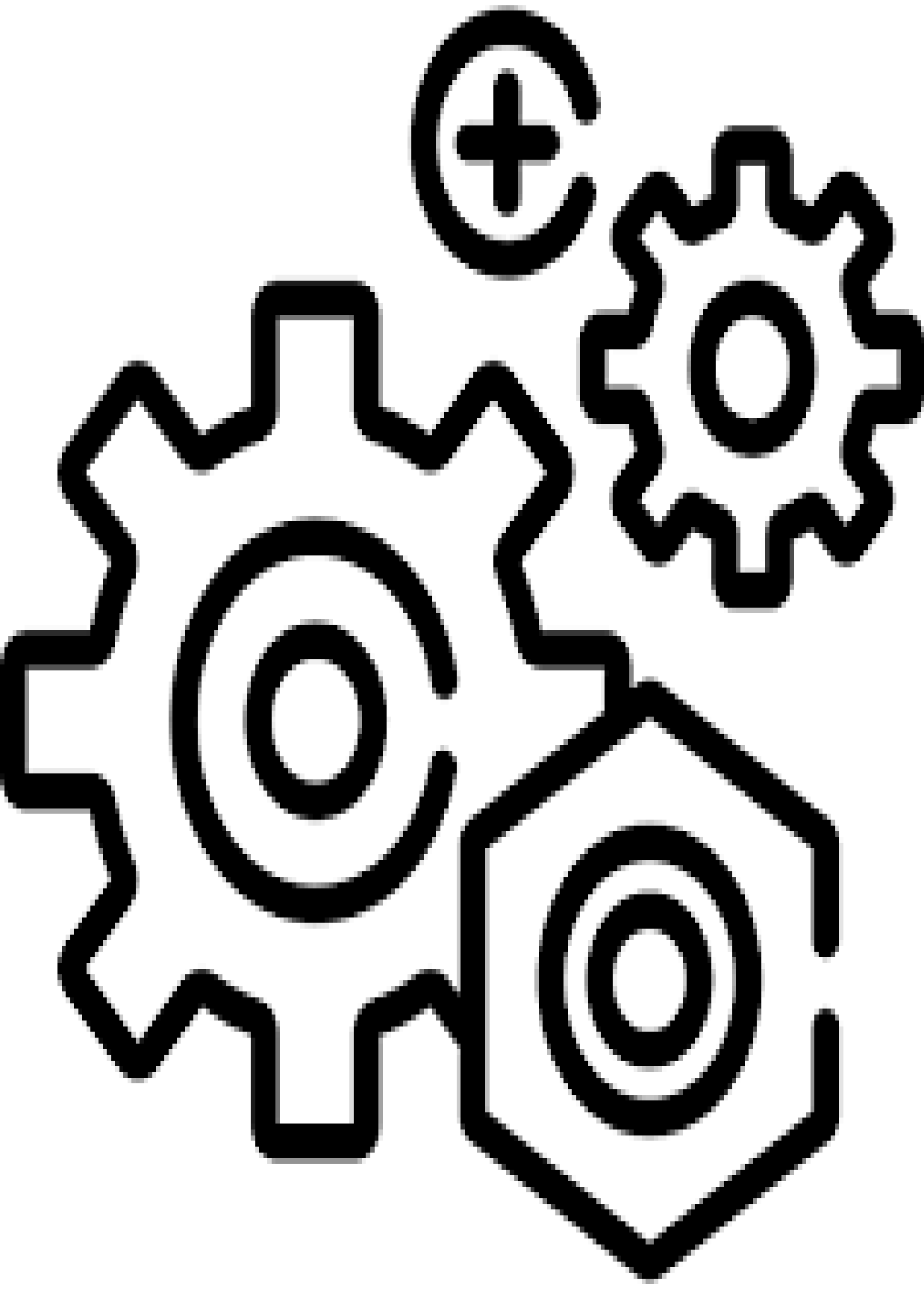
1	Visual Alerts Dashboard icon/text, such as “Eye Closed” or “ NO Driver” or "Take a break," for subtle warnings.
2	Audible Warnings Progressive tones (70-85 dB) on Buzzer for increased urgency.
3	Haptic Feedback Vibration or steering wheel shudder for direct physical alerts.
4	System Intervention Gentle speed reduction

Methodology



Circuit Design



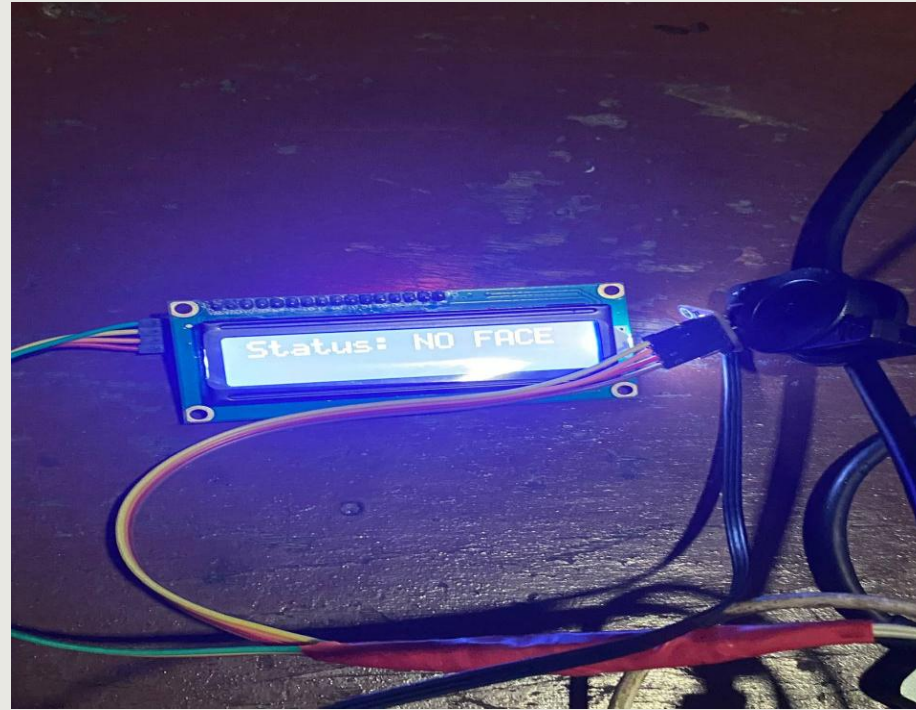
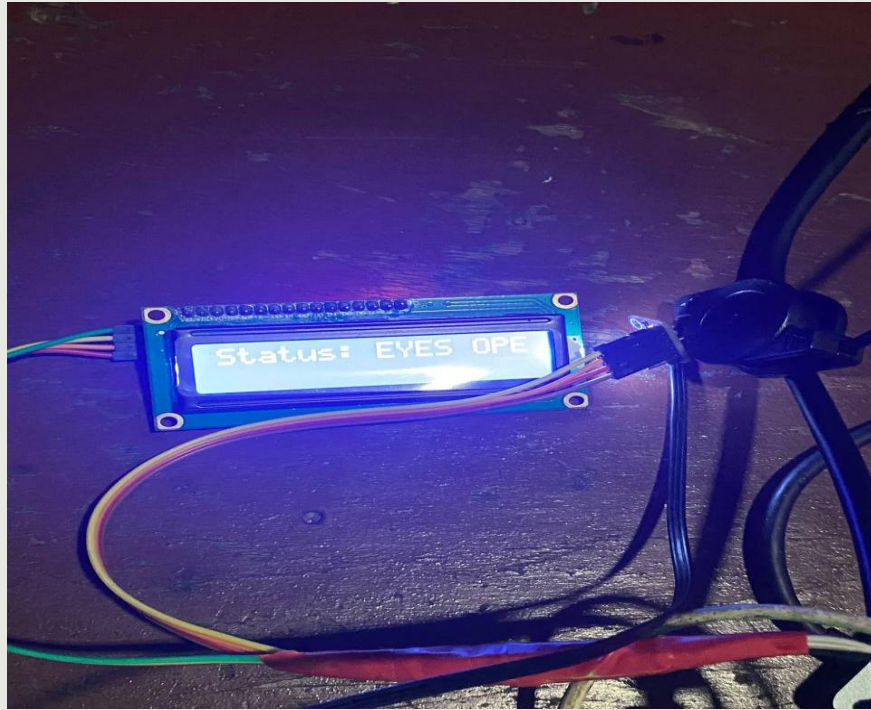


Component Description

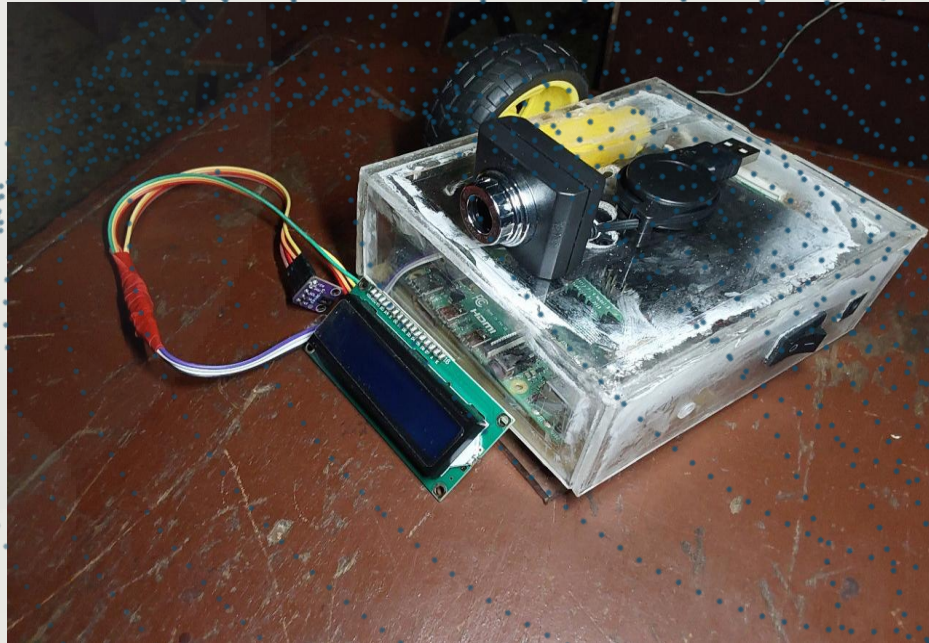
Main Components

- Battery: 12V, 18Ah Lithium-ion
- Switch
- Raspberry PI 4B
- Raspberry PI 4 USB Camera
- GY Max30102 Sensor
- Car Model motor
- Battery Level Indicator
- Small fan for System Aeration
- LCD: 1602 display for monitoring

Project Demonstration



Prototype Design



Automatic alerts and vehicle control mechanisms for safety.

Cost-effective and scalable solution suitable for personal and commercial vehicles.

Real-Time Processing: Achieves high accuracy (~80%) with minimal latency (<100ms inference, <10ms alert activation)





Challenges & Recommendations for Adoption

Challenges

- Driver acceptance. Sensor Intrusiveness versus Comfort
- Data privacy concerns.
- Sensor reliability in extreme conditions.
- False Positives

Recommendations

- Sensor Housing
- Conduct public education campaigns on drowsiness awareness.
- Develop cost-effective, adaptable aftermarket solutions.
- Remote Monitoring
- Conduct on-road trials to validate performance in varying condition



Conclusion: A Safer Future on the Roads

The developed system successfully detects driver drowsiness using a combination of facial and physiological indicators, providing timely alerts and automated speed control to prevent accidents. The prototype demonstrated robust performance under controlled conditions, achieving high detection accuracy and rapid response times. Integration of hardware and software components was achieved efficiently, validating the feasibility of a low-cost, non-intrusive solution for fatigue-related accident prevention



**THANK
YOU**